

Diminishing Returns of Healthcare Spending: A cross-country analysis of life expectancy in 2019

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Introduction

This report investigates the relationship between life expectancy at birth and healthcare spending across 59 countries in the year 2019. Health spending is one of the largest items in national budgets, ranging from 18–20% [2]. It is therefore important to understand whether high spending results in better health outcomes. This raises the question: **To what extent is the association between healthcare spending and life expectancy characterized by diminishing returns across countries?**

Data sample

I used data from Our World in Data. The dataset, *Life expectancy vs. health spending* [1], consisted of 18,809 rows and 6 columns; each row is one country in one year (1950–2025). It combines the following three indicators:

- **Life expectancy at birth** (years)
- **Health expenditure per capita** (int.-$)
- **World region**

To clean the data, I removed any rows that had missing values or non-positive spending, leaving only 2,047 rows (59 countries, 1970–2023). Moreover, I filtered the data to focus on the year 2019 because it was the last full year before the COVID-19 pandemic, giving $n = 59$ countries. This step avoided pandemic-related distortions.

Method

The analysis was written in Python (pandas, NumPy, SciPy, Matplotlib) and uses two tests.

1. Model comparison. To begin with, I created a scatter plot to visualize the relationship between life expectancy and health spending. The plot showed a curve that rises but gradually becomes flatter, suggesting diminishing returns. To test this, I fitted a linear model $\mathrm{LE} = a + b\,x$ and a logarithmic model $\mathrm{LE} = a + b\,\ln(x)$, where x is the spending per person. Then, to further compare how well each model represented the data, I calculated R^2.

2. Split-slope test. Secondly, I conducted a test that assumes no curve shape. Using the median healthcare spending ($2,645), the countries ($n = 59$) were divided into two groups, low-spending ($n = 30$) and high-spending ($n = 29$). To calculate the difference between the two slopes, I used the following formula:

$$z = \frac{b_{\text{low}} - b_{\text{high}}}{\sqrt{\sigma^2_{\text{low}} + \sigma^2_{\text{high}}}}. \tag{1}$$

Results and discussion

Figure 1 shows that life expectancy rises steeply at low spending; however, not at a constant rate. The data points and the logarithmic model follow similar trends, with the curve flattening above $5,000. Another reason why the logarithmic model is well suited for the data is its higher R^2 value of 0.62 compared with 0.46 for the linear model. This suggests that higher health spending levels are associated with smaller gains in life expectancy. Two clear negative outliers lie below the predicted curve. The United States has the highest health spending ($11,067 per person) out of the 59 countries, yet its life expectancy (78.9 years) is lower than the countries that spend a significantly lower amount. On the other hand, South Africa has a lower life expectancy (66.1 years) than the countries that spend similar amounts on healthcare. These outliers suggest that healthcare efficiency, access, and lifestyle matter as well as spending.

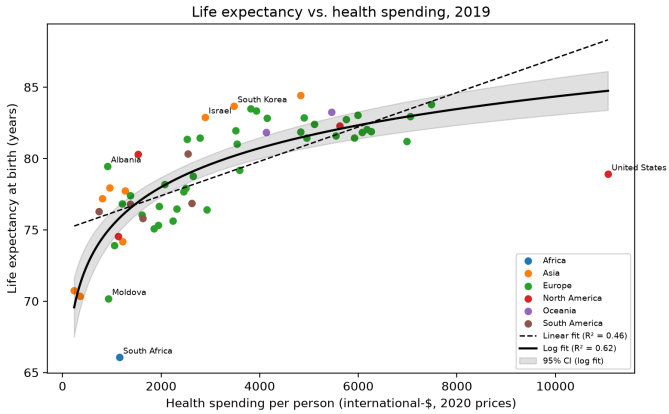


Figure 1: Plot displays life expectancy (y-axis) vs. health spending (x-axis) (2019). Each dot represents a country and is color-coded by region. The plot includes a linear and a log fit, with the grey band representing the 95% CI of the log fit.

Figure 2 confirms diminishing returns without assuming a curve shape. In low-spending countries, the slope is $+2.49 \pm 1.54$ per additional \$1,000 spent. This adds an additional 2.5 years of life expectancy. In contrast, the high-spending group's slope is -0.04 ± 0.37. While this value is very close to zero, it suggests no further gains in life expectancy. In addition, $z = 3.3$ provides strong statistical evidence that the association is significantly stronger at low spending countries when compared to high spending countries.

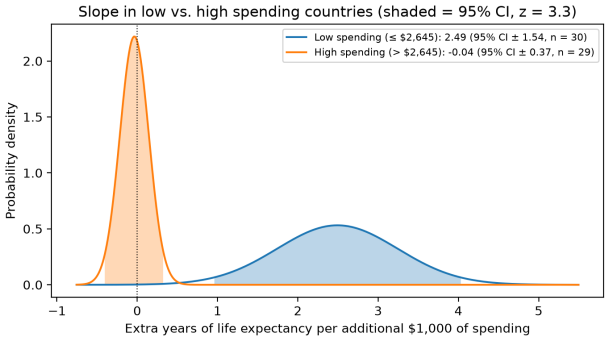


Figure 2: Two normal distribution curves representing low- and high-spending countries (2019). The dotted line indicates no association.

Conclusion

In conclusion, both analyses provide strong evidence of diminishing returns in 2019. In low-spending countries, an extra \$1,000 per person added 2.5 more years of life expectancy. Contrastingly, high-spending countries indicated no association with further gains in life expectancy. This means that the relationship becomes flatter as health spending increases.

References

[1] Our World in Data. *Life expectancy vs. health expenditure*. Data from UN World Population Prospects (2024) and OECD (2026). `https://ourworldindata.org/grapher/life-expectancy-vs-health-expenditure` (accessed 7 October 2026).

[2] Morgan, D., & Mueller, M. (2026). *Latest health spending trends and outlook*. OECD Health Working Papers. `https://doi.org/10.1787/9fc8d4b1-en`